

Sec 5.1 #8

Thm. Let V be a finite vector space, $T: V \rightarrow V$ be a linear operator.

If β, γ are distinct ordered basis of V ,

then

$$\det([T]_{\beta}) = \det([T]_{\gamma}).$$

Proof. Since

$$\det([T]_{\beta}) = \underbrace{\det([T]_{\beta}^{\gamma})}_{\text{inverse}} \cdot \det([T]_{\beta}^{\beta}) \cdot \underbrace{\det([T]_{\gamma}^{\beta})}_{\text{inverse}}$$

$$= \det([T]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [T]_{\gamma}^{\beta}) = \det([T]_{\gamma}^{\gamma})$$

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \phi_{\alpha} \downarrow & G & \downarrow \phi_{\beta} \\ F^n & \xrightarrow{[T]_{\alpha}^{\beta}} & F^n \end{array}$$

Sec 5.1 #9

Thm. Let $T: V \rightarrow V$ be a linear operator on finite v.s V .

T is invertible iff every eigenvalue of T is nonzero.

Proof. There exists a zero eigenvalue

$$\Leftrightarrow \exists v \neq 0 \text{ s.t. } T(v) = 0 \cdot v = 0$$

$$\Leftrightarrow \ker T \neq \{0\}$$

$$\Leftrightarrow T \text{ is not injective.}$$

Thm. Let $T: V \rightarrow V$ be an invertible linear operator on finite v.s V .

λ is eigenvalue of T iff λ^{-1} is eigenvalue of T^{-1} .

Proof. λ is eige. of T

$$\Leftrightarrow \exists v \neq 0 \text{ s.t. } T(v) = \lambda v$$

$$\text{And, } T(v) = \lambda v$$

$$\Leftrightarrow (T^{-1} \circ T)(v) = T^{-1}(\lambda v) = \lambda \cdot T^{-1}(v)$$

$$\Leftrightarrow \lambda^{-1} v = T^{-1}(v)$$

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observation: Let V be fin. v.s., $I_V: V \rightarrow V$ be a identity map

$$\det(\lambda I_V - t I_V) = \det((\lambda - t)I) = (\lambda - t)^n$$

Thm. If A is diagonalizable matrix with a unique eigenvalue,
then $A = \lambda I$ where λ is eigenvalue of A

Proof. Since A is diagonalizable, and has only one eigenvalue,

$$\begin{aligned} A &= Q^{-1} D Q \quad \text{where } D = \lambda I \\ &= Q^{-1} Q \lambda I \\ &= \lambda I \end{aligned}$$

Example

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not diagonalizable;

$\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1-t)^2$, thus has eigenvalue as only 1.

But, if $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is diagonalizable, it is scalar matrix.

This is contradiction.

Thm. If a matrix A is similar to B , then $\det A = \det B$.

Proof. $\det A = \det Q B Q^{-1} = \det Q^{-1} \det Q \det B = \det B$.

Thm. characteristic of linear operator does not depend on a basis.

Proof. Since $A = Q^{-1} B Q \Rightarrow A - I = Q^{-1} B Q - I$
 $= Q^{-1} (B - I) Q$

sec 5.1. #14

Thm. Let $T: V \rightarrow V$ be a linear operator on finite vec. sp. over F .

Put $A = [T]_{\beta}$.

1. For $v \in V$, if $\phi_{\beta}(v)$ is eigenvector of A corresponding to λ ,
then v is eigenvector of T corresponding to λ .

2. If $\lambda \in F$ is eigenvalue of A and T , then

$v \in F^n$ is eigenvector of A corresponding to λ iff $\phi_{\beta}^{-1}(v)$ is eigenvector of T corresponding to λ .

Proof. 1.

$$\phi_{\beta}(T(v)) = [T(v)]_{\beta} = [T]_{\beta} [v]_{\beta} = A \cdot \phi_{\beta}(v) = \lambda \phi_{\beta}(v) = \phi_{\beta}(\lambda v)$$

Since ϕ_{β} is isomorphism, $T(v) = \lambda v$.

2.

$$\begin{aligned} \lambda \phi_{\beta}^{-1}(v) &= \phi_{\beta}^{-1}(\lambda v) = \phi_{\beta}^{-1}(A v) = \phi_{\beta}^{-1}([T]_{\beta} v) = \phi_{\beta}^{-1}([T]_{\beta} [\phi_{\beta}^{-1}(v)]_{\beta}) \\ &= \phi_{\beta}^{-1}([T(\phi_{\beta}^{-1}(v))]_{\beta}) \\ &= T(\phi_{\beta}^{-1}(v)) \end{aligned}$$

Thm. (λ, x) is eigen- of $T: V \rightarrow V$. Then for all $n \in \mathbb{N}$, (λ^n, x) is eigen- of T^n .

Proof. $T^n(x) = T^{n-1}(\lambda x) = \lambda \cdot T^{n-1}(x) = \dots = \lambda^n x$.

Q.E.D.

sec 5.1 #19

Thm. Let $T: V \rightarrow V$ linear, $\dim V < \infty$. $c \in F$ be a scalar.

If $\lambda \in F$ is eigenvalue of T corresponding to $v \in V$
 then $\lambda - c$ is eigenvalue of $T - cI$ corresponding to $v \in V$. Moreover,
 T is diagonalizable iff $T - cI$ diagonalizable.

Proof. $T(v) = \lambda v$ implies

$$(T - cI)(v) = T(v) - cv = (\lambda - c)v.$$

Sec 5.1 #18

Thm. Let $T: M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R}) : A \mapsto A^t$.

1. Eigenvalue of T is only ± 1 .

2.

Proof. Let $\lambda \in \mathbb{R}$ be an eigenvalue of T corresponding to A . Then,

$$T(A) = \lambda A = A^t$$

$$\lambda A = A^t \Leftrightarrow \lambda A^t = A$$

$$\text{Thus, } \lambda A = \lambda^t A \Rightarrow (\lambda^2 - 1)A = 0. \text{ Since } A \neq 0, \lambda^2 = 1.$$

Let β be a basis of $M_n(\mathbb{R})$.

$[T]_\beta$ is diagonal iff $\forall A \in \beta, T(A) = \pm A^t$.

$$\beta = \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right\} \leftarrow n \text{ elements}$$

$$\cup \left\{ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \dots \right\} \leftarrow \frac{n(n-1)}{2}$$

$$\cup \left\{ \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \dots \right\}$$

) n^2 elements

Sec 5.1 #19
Thm. $A, B \in M_{nn}(\mathbb{C})$.

For an invertible B , there exists $c \in \mathbb{C}$ s.t. $A + cB$ is not invertible.

Proof. $A + cB$ is not invertible

$$\Leftrightarrow \det(A + cB) = 0 \Leftrightarrow \det(B^{-1}A + cI) = 0$$

$$\Leftrightarrow -c \text{ is eigenvalue of } B^{-1}A.$$

Since field is set by $\mathbb{C}$, $B^{-1}A$ always has a eigenvalue λ .
nam, taking $c = -\lambda$.

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

Thm. Let $A \in M_{nn}(F)$, $f(t) = \det(A - tI)$.

Then, $f(0) = \det A$. More precisely,

$$f(t) = (-1)^n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + \det A.$$

And, for some polynomial $\deg g(t) \leq n-2$

$$f(t) = (A_{11} - t)(A_{22} - t) \dots (A_{nn} - t) + g(t).$$

And, $\text{tr}(A) = (-1)^{n-1} a_{n-1}$.

Proof. Using induction.

For $n=2$, $f(t) = (a-t)(d-t) - bc = (A_{11} - t)(A_{22} - t) - bc$

Suppose for $n \in \mathbb{N}$, the statement is true.

Then, for $A = \begin{pmatrix} B & \\ & A_{n+1,n+1} \end{pmatrix}$, $\det(A - tI) = \det(B - tI) \cdot (A_{n+1,n+1} - tI)$

Now, $f(t) = (A_{11} - t)(A_{22} - t) \dots (A_{nn} - t) + g(t)$

$$= (-1)^n t^n + (-1)^{n-1} (A_{11} + A_{22} + \dots + A_{nn}) t^{n-1} + g(t) + \dots$$

Thm. Let $T: V \rightarrow V$ be a linear operator, $g(t) \in F[t]$.

If x is eigenvector of T corresponding to λ , then $g(T)(x) = g(\lambda)x$.

$$\text{Proof. } \left(\sum_{i=0}^n a_i T^i \right)(x) = \sum_{i=0}^n a_i T^i(x) = \sum_{i=0}^n a_i \lambda^i x$$

Thm. If a linear operator $T: V \rightarrow V$ is diagonalizable and $f(t)$ is characteristic poly. of T , then $f(T) = 0$.

Proof. Let $x \in V$ be given. Then, $x = \sum_{i=1}^n a_i x_i$ where x_i is eigenvector corresponding to λ_i

$$\begin{array}{ccc} \text{def of sum of functions} & T(x) = \lambda x \Rightarrow T(ax) = a\lambda x & \\ \downarrow & \downarrow & \end{array}$$

$$f(T) \left(\sum_{i=1}^n a_i x_i \right) = \sum_{i=1}^n f(T)(a_i x_i) = \sum_{i=1}^n a_i \underbrace{f(\lambda_i)}_{=0}(x_i) = \sum_{i=1}^n 0 = 0.$$

Thm. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be ^{distinct} eigenvalues of $T: V \rightarrow V$.

And, for each $1 \leq i \leq k$, define S_i is a basis of E_{λ_i} .

Then, $\bigcup_{i=1}^k S_i$ is linearly independent.

Proof. Using induction. For $k=1$, trivial.

Suppose that for $n \in \mathbb{N}$, the statement is true.

Denote $S_i = \{V_{i,1}, V_{i,2}, \dots, V_{i,n_i}\}$. Let

$$\sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot V_{ij} = 0.$$

Taking $T - \lambda_{n+1}I$ for both side, then

$$\begin{aligned} & (T - \lambda_{n+1}I) \left(\sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot V_{ij} \right) \\ &= \sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} (\lambda_i - \lambda_{n+1}) \cdot V_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^{n_i} a_{ij} (\lambda_i - \lambda_{n+1}) V_{ij} \end{aligned}$$

By assumption, this implies $\forall 1 \leq i \leq n, \forall 1 \leq j \leq n_i, a_{ij} (\lambda_i - \lambda_{n+1}) = 0$.

Since $\lambda_i \neq \lambda_{n+1} \quad \forall 1 \leq i \leq n$, $a_{ij} = 0$.

$$\text{Now, } \sum_{i=1}^{n+1} \sum_{j=1}^{n_i} a_{ij} \cdot V_{ij} = \sum_{j=1}^{n_{n+1}} a_{n+1,j} \cdot V_{n+1,j} =$$

Meanwhile, S_{n+1} is linearly independent, $\forall 1 \leq j \leq n_{n+1} \quad a_{n+1,j} = 0$.

Thm. Let $T: V \rightarrow V$ be a linear map on finite vector space V ,
and λ be an eigenvalue with multiplicity m . Then,

$$1 \leq \dim E_\lambda \leq m$$

Proof. Using replacement Theorem, Let basis of E_λ as $\{v_1, \dots, v_p\}$ and basis of V as $\beta = \{v_1, \dots, v_p, \dots, v_n\}$

Then,

$$[T]_\beta = \begin{pmatrix} \lambda I_p & B \\ 0 & C \end{pmatrix}.$$

Now, $f(t) = \det([T]_\beta - tI) = \det \begin{pmatrix} (\lambda - t)I_p & B \\ 0 & C - tI \end{pmatrix} = \det((\lambda - t)I_p) \cdot \det(C - tI) = (\lambda - t)^p g(t)$

Therefore, $\dim E_\lambda = p \leq \deg f(t) = m$

^TThm. Let $T: V \rightarrow V$ be a linear operator on finite vector space V .

T is diagonalizable iff:

- 1). Characteristic polynomial of T splits over F .
- 2). For all eigenvalue λ , multiplicity of $\lambda = \dim E_{\lambda}$.

Proof. Suppose that T is diagonalizable. Put $\beta = \{v_1, \dots, v_n\}$ be a basis s.t. $[T]_{\beta}$ is diag.

$T(v_i) = \lambda_i v_i$ for some $\lambda_i \in F$, since $[T]_{\beta}$ is diagonal.

And, the characteristic polynomial of T :

$$f(t) = \prod_{i=1}^n (\lambda_i - t).$$

Thus, $\dim E_{\lambda_i} = \text{mult- of } \lambda_i$.

Conversely, suppose that

Thm. Let V be a finite vector space, and $W_1, W_2, \dots, W_k \leq V$ be subspaces.

TFAE:

1) $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$

2) $V = \sum_{i=1}^k W_k$ and if $\sum_{i=1}^k V_k = 0$ where $V_i \in W_i$, then $V_1 = V_2 = \dots = V_k = 0$.

3) For any $v \in V$, there exists a unique representation s.t.

$$v = V_1 + \dots + V_k \quad \text{where } V_i \in W_i.$$

4) For basis γ_i of W_i , $\bigcup_{i=1}^k \gamma_i$ is basis of V .

5) For each $1 \leq i \leq k$, there exist a basis γ_i of W_i s.t.

$$\bigcup_{i=1}^k \gamma_i \text{ is basis of } V.$$

Thm.

A linear operator $T: V \rightarrow V$ is diagonalizable

iff

V is direct sum of eigenspace of T .

Thm. Let $T: V \rightarrow V$ be a linear operator on finite vector space V .

If there exists a basis β of V s.t. $[T]_\beta$ is upper-triangular matrix, then characteristic polynomial of T splits over F .

Proof. Using induction. For $\dim V = 2$, trivial.

Suppose that for $\dim V = n \in \mathbb{N}$, the statement is true.

Let $T: F^{n+1} \rightarrow F^{n+1}$ be a linear operator, and β be a basis of F^{n+1} s.t. $[T]_\beta$ is upper-tri.

Write $\beta = \{v_1, \dots, v_n, v_{n+1}\}$. Then, for a restriction

$$T|_{F^n}: F^n \rightarrow F^n; x \mapsto (T_1 \circ T(x), T_2 \circ T(x), \dots, T_n \circ T(x))$$

satisfies: $[T|_{F^n}]_{\beta \setminus \{v_{n+1}\}}$ is upper-triangular,

since for all $1 \leq i \leq n$, $(T_{n+1} \circ T)(v_i) = 0$, $(T|_{F^n}(v_i), 0) = T(v_i)$.

Now,

$$\det(T - tI) = \det \begin{pmatrix} T|_{F^n} - tI & \vdots \\ 0 & \dots & 0 & \lambda - t \end{pmatrix} = (\lambda - t) \cdot \det(T|_{F^n} - tI) = f(t)$$

And, by assumption, $\det(T|_{F^n} - tI)$ splits over F , so is $f(t)$.

Thm. Let $T: V \rightarrow V$ be a linear operator on $V \cong F^n$.

$\lambda_1, \dots, \lambda_k$ be distinct eigenvalues with multiplicity m_i .

If there exists a basis of V s.t. $[T]_\beta$ is upper-triangular,

then

$$[T]_\beta = \begin{pmatrix} \lambda_1 I_{m_1} & & 0 \\ & \ddots & \\ 0 & & \lambda_k I_{m_k} \end{pmatrix} + \begin{pmatrix} 0 & & \vdots \\ & \ddots & \\ 0 & & 0 \end{pmatrix}$$

Proof. Since $[T]_\beta$ is upper-triangular, $\det([T]_\beta - tI) = \prod_{i=1}^n (A_{ii} - t)$, where $A = [T]_\beta$.

And, by assumption, $f(t) = (\lambda_1 - t)^{m_1} \dots (\lambda_k - t)^{m_k} = \prod_{i=1}^n (A_{ii} - t)$.

Thus proved.

Thm. Let $A \in M_n(F)$ be a matrix which is similar to an upper triangle,
and $\lambda_1, \dots, \lambda_k$ be distinct eigenvalue of A with multiplicity m_i .

$$1) \operatorname{tr} A = \sum_{i=1}^k m_i \lambda_i$$

$$2) \det A = \prod_{i=1}^k \lambda_i^{m_i}.$$

Proof. Trivial:

Thm. If a characteristic polynomial of $A \in M_n(F)$ splits over F , then A is similar to an upper triangle matrix.

Proof. For $n=1$, trivial.

Suppose that for $n-1 \leq n$, the statement is true.

Let $A \in M_n(F)$, which is characteristic polynomial splits.

Denote

$$A = \begin{pmatrix} A' & \vdots \\ \dots & a_n \end{pmatrix} \quad \text{where } A' \in M_{n-1}(F).$$

Since $f_A(t)$ splits over F , there is an eigenvector $v_n \in V$.

Let $\beta = \{v_1, \dots, v_n\}$ be a basis of F^n .

Put $P = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$. Then,

$$P^{-1}AP = \begin{pmatrix} P^{-1}Av_1 & \dots & P^{-1}Av_n \end{pmatrix} = \begin{pmatrix} P^{-1}Av_1 & \dots & P^{-1}Av_n \end{pmatrix}$$

⌋

Thm. Let $T: V \rightarrow V$ be an invertible linear operator. with $\dim V = n$.

If $\lambda \in F$ is eigenvalue of T , then λ^{-1} is eigenvalue of T^{-1} .

Precisely, $\{v \in V \mid T(v) = \lambda v\} = \{v \in V \mid T^{-1}(v) = \lambda^{-1}v\}$.

Moreover, T is diagonalizable iff T^{-1} is diagonalizable.

Proof. Since $T(v) = \lambda v$ for some non-zero vector v ,

$$\begin{aligned} T(v) = \lambda v &\Leftrightarrow v = (T^{-1} \circ T)(v) = T^{-1}(\lambda v) = \lambda T^{-1}(v) \\ &\Leftrightarrow T^{-1}(v) = \lambda^{-1}v. \end{aligned}$$

Suppose that T is diagonalizable. Then, there exists a basis β of V s.t.

$$[T]_{\beta} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}, \text{ which is diagonal.}$$

$$\text{Now, } [T^{-1}]_{\beta} = [T]_{\beta}^{-1} = \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix}$$

Thm. Let $A \in M_n(F)$ be given. Then, A and A^t has same characteristic polynomial.

That is, A and A^t has same eigenvalue with same multiplicity.

Denote E_{λ} is eigenspace of A corresponding to λ ,

$$E_{\lambda} \text{ is } \ll A^t \ll \lambda.$$

For any eigenvalue λ , $\dim E_{\lambda} = \dim E_{\lambda}^t$.

Moreover, A is diagonalizable iff A^t is diagonalizable.

Proof.

Def. Let V be a finite vector space $T, U: V \rightarrow V$ be linear operators.

If there exists a basis β s.t

$[T]_\beta, [U]_\beta$ are diagonal Matrix.

Then, it is said to be T and U is simultaneously diagonalizable.

In Matrix,

$\exists Q \in M_{n \times n}$ s.t $\begin{cases} Q^T A Q \\ Q^T B Q \end{cases}$ are diagonal.

Thm. If $T, U: V \rightarrow V$ are simultaneously diagonalizable,
then T, U is commute. ($TU = UT$).

Proof. Since diagonal Matrices are commute,

$$[T \circ U]_\beta = [T]_\beta [U]_\beta = [U]_\beta [T]_\beta = [U \circ T]_\beta.$$

Thm. If T is diagonalizable,

then for all $n \in \mathbb{N}$, T, T^n are simul-diagonal-.

Proof. $[T^n]_\beta = [T]_\beta^n$ is diagonal, especially

$$= \begin{pmatrix} \lambda_1^n & & 0 \\ & \lambda_2^n & \\ 0 & & \ddots \\ & & & \lambda_k^n \end{pmatrix}.$$

Thm. Let $T: V \rightarrow V$ be a linear operator, and $W \subseteq V$.

W is called T -invariant subspace of V if:

$$T[W] \subseteq W.$$

And, for $x \in V$,

$$W = \langle \{T^n(x) \mid n=0, \dots\} \rangle$$

is called T -cyclic subspace generated by x .

Thm. Let $T: V \rightarrow V$ be a linear operator on fin. v.s V ,

$W \subseteq V$ be T -invariant subspace. Then,

f_{T_W} divides f_T .

Proof. Let $\beta = \{v_1, \dots, v_k\}$ be a basis of W , and

$\bar{\beta} = \{v_1, \dots, v_k, \dots, v_n\}$ be a basis of V .

$$\text{Then, } [T]_{\bar{\beta}} = \begin{pmatrix} [T_W]_{\beta} & \vdots & \cdot \\ \hline 0 & \vdots & \cdot \end{pmatrix}$$

because for all $1 \leq i \leq k$, $T(v_i) = \sum_{j=1}^k a_{ij} v_j$ for some $a_{ij} \in F$.

Now, a characteristic polynomial of T :

$$\det([T]_{\bar{\beta}} - tI) = \det([T_W]_{\beta} - tI) \cdot \det(\cdot - tI)$$

That is, f_{T_W} divides f_T .

Thm. Let $T: V \rightarrow V$ be a linear operator on fin. v.s. V ,

And, $W = \langle \{T^n(v) \mid n=0,1,\dots\} \rangle$ where $v \neq 0$.

Put $k = \dim W$

$$\det \begin{pmatrix} -t & & & & \\ 1 & -t & & & \\ 0 & 1 & -t & & \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & -t \end{pmatrix} = (-1)^{k+k} - 1$$

1). $\{v, T(v), \dots, T^{k-1}(v)\}$ is basis of W

2). If $a_0 v + a_1 T(v) + \dots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$,

then $f_W(t) = (-1)^k (a_0 + a_1 t + \dots + a_{k-1} t^{k-1} + t^k)$

Proof. Let $j \in \mathbb{N}$ be the largest number s.t

$$\beta = \{v, T(v), T^2(v), \dots, T^{j-1}(v)\}$$

is linearly independent. Put $Z = \langle \beta \rangle$. Then, β is basis of Z .

Moreover, $\beta \cup \{T(v)\}$ is not linearly independent,

$$T^j(v) = \sum_{i=0}^{j-1} b_i T^i(v) \in Z$$

Now, for any $w = \sum_{i=0}^{j-1} b_i T^i(v) \in Z$,

$$T(w) = T\left(\sum_{i=0}^{j-1} b_i T^i(v)\right) = \sum_{i=0}^{j-1} b_i T^{i+1}(v) \in Z.$$

Since lemma $W \subseteq Z$, and $Z \subseteq W$, because β is subset of spanning set of W .

To prove 2), if $a_0 v + a_1 T(v) + \dots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$, then

$$[T^k]_{\beta} = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{k-1} \end{pmatrix}, \text{ thus } f_W(t) = (-1)^k (a_0 + a_1 t + \dots + a_{k-1} t^{k-1} + t^k).$$

Lemma. Let $T: V \rightarrow V$ be linear, W be a T -cyclic subspace generated by $v \neq 0$.

Then, W is the smallest T -invariant subspace containing $v \in V$.

Proof. Let $x \in W$ be given, then, x can be represented by

$$x = \sum_{i=0}^n \alpha_i T^i(v)$$

$$\text{Now, } T(x) = \sum_{i=0}^n \alpha_i T^{i+1}(v) \in W.$$

And, suppose that $A \subseteq V$ is T -invariant. containing $v \in V$.

Then, $\{v, T(v), T^2(v), \dots\} \subseteq A$, thus $W \subseteq A$.

$$\begin{pmatrix} -t & 0 & 0 & \dots & 1 & -a_0 \\ 1 & -t & & & & -a_1 \\ 0 & 1 & & & & \\ \vdots & 0 & \ddots & & & \\ \vdots & \vdots & & \ddots & & \\ 0 & 0 & \dots & 0 & 1 & -a_{n-1}-t \end{pmatrix} = A_n.$$

$$\begin{pmatrix} -t & 0 & & & \\ 1 & -t & & & \\ 0 & 1 & & & \\ \vdots & 0 & \ddots & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \dots & 0 & 1-t \end{pmatrix} = \underline{B_n}.$$

$$\underline{\det B_n = (-1)^n t^n.}$$

$$\begin{aligned} \det A_n &= \underline{(-a_{n-1}-t) \cdot \det B_{n-1}} \\ &\quad + a_{n-2} \cdot \det B_{n-2} \\ &\quad + a_{n-3} \det B_{n-3} \\ &\quad \vdots \\ &\quad + a_0 \det B_0 \end{aligned}$$

5.4 #3.

$$\forall v \in E_\lambda, T(v) = \lambda v \in E_\lambda.$$

5.4 #4.

If $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, for any $g(t) \in F[t]$.

proof. Let $g(t) = \sum_{i=0}^n a_i t^i$. Then, for any $x \in W$,

$$\begin{aligned} g(T)(x) &= \left(\sum_{i=0}^n a_i T^i \right) (x) \\ &= \sum_{i=0}^n a_i T^i(x) \in W, \text{ because } T^i(x) \in W. \end{aligned}$$

5.4 #5.

Let $W_1, W_2 \subseteq V$ be T -invariant.

for any $x \in W_1 \cap W_2$, $T(x) \in W_1$ and $T(x) \in W_2$. $\therefore T(x) \in W_1 \cap W_2$.

5.4 #13, 14.

Let $T: V \rightarrow V$ be a linear operator, W be a T -cyclic subspace generated by $v \in V$.

For any $x \in W$,

$$x \in W \iff \exists g(t) \in F[t] \text{ s.t. } x = g(T)(v).$$

$$\text{proof. } x \in W \iff x = \sum_{i=0}^n a_i T^i(v) \text{ for some } n \in \mathbb{N}, a_i \in F.$$

Since for $k = \dim W$, $\{T^0(v), \dots, T^{k-1}(v)\}$ is basis of W , clear.

Sec 5.4 #17

For $A \in M_{n \times n}$, $\dim(\langle \{I, A, A^2, \dots\} \rangle) \leq n$.

Proof. Let $f(t) = \det(A - tI)$. Then, $f(A) = 0$.

Since $\deg f(t) = n$, $\{I, A, \dots, A^n\}$ is linearly dependent.

Sec 5.4 #18.

Let $A \in M_{n \times n}(F)$ with char. poly.

$$f(t) = (-1)^n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0.$$

Clearly, $a_0 = \det A$.

Sec 5.4 #20.

Let $T: V \rightarrow V$, and $V = \langle \{T^n(v) \mid n=0, 1, \dots\} \rangle$ for some $v \in V$. Then,

A linear map $U: V \rightarrow V$ satisfies $T \circ U = U \circ T$
iff

$$U = g(T) \text{ for some } g(t) \in F[t]$$

Proof. $\Leftarrow$ is trivial.

$\Rightarrow$) Since $U(v) \in V$, $U(v) = g(T)(v)$ for some $g(t) \in F[t]$.

Now, for any $x = \sum a_i T^i(v) \in V$,

$$\begin{aligned} U(x) &= U\left(\sum a_i T^i(v)\right) = \sum a_i T^i(U(v)) \\ &= \sum a_i T^i(g(T)(v)) \\ &= g(T)\left(\sum a_i T^i(v)\right) \\ &= g(T)(x). \quad \square \end{aligned}$$

#21, 22 is trivial by #20.

Sec 5.4 #23.

$T: V \rightarrow V$, W T -invariant, $\lambda_1, \dots, \lambda_k$ distinct eigenvalues. Corresponding v_i .

$$v_1 + v_2 + \dots + v_k \in W \quad \text{iff} \quad v_1, v_2, \dots, v_k \in W.$$

Proof. For $k=1$, trivial.

For $k \in \mathbb{N}$, suppose it is true.

Now, for $v_1, \dots, v_{k+1}$, suppose:

$$v_1 + v_2 + \dots + v_{k+1} \in W.$$

$$\text{Then, } \sum_{i=1}^{k+1} T(v_i) \in W, \text{ and}$$

$$= \sum_{i=1}^{k+1} \lambda_i v_i.$$

If $\lambda_{k+1} \neq 0$, then $\lambda_{k+1}(\sum_{i=1}^{k+1} v_i) \in W$, this implies

$$\sum_{i=1}^{k+1} \lambda_i v_i = \lambda_{k+1} \left(\sum_{i=1}^{k+1} v_i \right)$$

$$= \sum_{i=1}^{k+1} (\lambda_i - \lambda_{k+1}) v_i$$

$$= \sum_{i=1}^k (\lambda_i - \lambda_{k+1}) v_i \in W \Rightarrow v_1, \dots, v_k \in W,$$

Thus $v_{k+1} \in W$.

If $\lambda_{k+1} = 0$, then clearly $v_{k+1} \in W$, obtain same conclusion.

Sec 5.4 #24. ?

Let $T: V \rightarrow V$ be a diagonalizable. Then, for any T -invariant W ,

$T|_W$ is diagonalizable.

Proof. Since T is diagonalizable, $V = \bigoplus_{i=1}^n E_{\lambda_i}$, where λ_i is distinct eigenvalue of T .

Now, for any $x \in W$, $x = \sum_{i=1}^n a_i v_i$ as $a_i \in F$, $v_i \in E_{\lambda_i}$, and unique representation.

Since above Lemma, all $v_i \in W$. That is,

$$W = \bigoplus_{i=1}^n (E_{\lambda_i} \cap W)$$

Sec 5.4 #25.

Let $T: V \rightarrow V$ be diagonalizable.

If $T \circ U = U \circ T$, then T, U simultaneously diagonalizable.

Proof. Let $\lambda \in \mathbb{F}$ be a non-zero eigenvalue of T . For $E_\lambda = \{v \in V \mid T(v) = \lambda v\}$,

Let $x \in E_\lambda$. Then, $U(x) = U(\lambda^{-1} T(x)) = \lambda^{-1} (T \circ U)(x)$

$\Leftrightarrow \lambda U(x) = T(U(x))$. That is, $U(x) \in E_\lambda$.

Step 1

If $\lambda = 0$, then $E_\lambda = \ker T$.

Let $x \in \ker T$. Then, $T(U(x)) = U(T(x)) = 0$. $\therefore U(x) \in \ker T$.

In sum, E_λ is U -invariant.

Now, $U|_{E_\lambda}$ is diagonalizable, but β_λ is basis of E_λ s.t. $[U|_{E_\lambda}]_{\beta_\lambda}$ is diagonal.

$\beta = \bigcup_\lambda \beta_\lambda$ is basis of V consist. by eigenvectors of T .

and $[U]_\beta$ is diagonal.

T diagonalizable

$$\Rightarrow V = E_{\lambda_1} \oplus \dots \oplus E_{\lambda_k}$$

$$\begin{array}{ccc} \text{span} \uparrow & \uparrow & \uparrow \\ \beta & = \beta_1 \cup \dots \cup \beta_k. \end{array}$$

Sec 5.4, #26.

Let $T: F^n \rightarrow F^n$ be a linear, having n numbers distinct eigenvalues.

Then, F^n is T -cyclic space.

proof. Let $\lambda_1, \dots, \lambda_n$ be eigenvalues of T .

Let v_i be eigenvector corresponding to λ_i .

Let $V = \sum_{i=1}^n v_i$, $W = \langle \{v_i, T(v_i), \dots\} \rangle$.

Then, W must contains of v_i s, thus $W = \bigoplus E_{\lambda_i} = V$.

$\Gamma T: V \rightarrow V, W: T\text{-invariant}$

$\bar{T}: V/W \rightarrow V/W; v+W \mapsto T(v)+W.$

$\eta: V \rightarrow V/W; v \mapsto v+W.$

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \eta \downarrow & & \downarrow \eta \\ V/W & \xrightarrow{\bar{T}} & V/W. \end{array}$$

$\Gamma_{\text{thm.}} f_T(t) = f_{\bar{T}}(t) \cdot f_{T_W}(t).$

Proof. Let $\gamma = \{v_1, \dots, v_k\}$ be a basis of W ,
 $\beta = \{v_1, \dots, v_k, \dots, v_n\}$ be a basis of V .

Then, $\alpha = \{v_{k+1}+W, \dots, v_n+W\}$ is basis of V/W .

Now,

$$[T]_{\beta} = \begin{pmatrix} [T]_{\gamma} & \vdots & \dots \\ 0 & [\bar{T}]_{\alpha} \end{pmatrix}.$$

[Sec 5.4 #32.

Let $T: V \rightarrow V$ be linear operator, $W_1, \dots, W_k$ be T -invariant subspace.

Then, $W_1 + \dots + W_k$ is T -invariant subspace.

Proof. Let $x_1 + \dots + x_k \in \sum_{i=1}^k W_i$ be given. Then,

$$T(x_1 + \dots + x_k) = \sum_{i=1}^k T(x_i) \in \sum_{i=1}^k W_i, \text{ since each } T(x_i) \in W_i.$$

[Sec 5.4 #33.

V is finite-dimensional.

Let $T: V \rightarrow V$ be a linear, $V = W_1 \oplus W_2$ for some T -invariant W_1, W_2 .

Let β_1, β_2 be basis of W_1, W_2 , respectively. Put $\beta = \beta_1 \cup \beta_2$.

Then,

$$[T]_{\beta} = [T_{W_1}]_{\beta_1} \oplus [T_{W_2}]_{\beta_2}.$$

Proof. Since V is direct sum of W_1 and W_2 , $\beta_1 \cap \beta_2 = \emptyset$.

Denote $\beta_1 = \{v_1, \dots, v_n\}$, $\beta_2 = \{w_1, \dots, w_m\}$.

$$T(v_i) = T_{W_1}(v_i) \in W_1, \quad T(w_i) = T_{W_2}(w_i) \in W_2.$$

Sec 5.4 #35.

Let $T: V \rightarrow V$ be a linear operator on finite dimensional V .

T is diagonalizable iff $V = \bigoplus_{i=1}^n W_i$ where $\dim W_i = 1$,
 W_i is T -invariant.

Proof. Suppose that T is diagonalizable. Then,

There exists a basis β s.t. $[T]_{\beta}$ is diagonal.

Then, for each $v_i \in \beta$, $\langle v_i \rangle$ is T -invariant, since

$$T(av_i) = aTv_i \in \langle v_i \rangle, \text{ for any } a \in F.$$

Now, $V = \bigoplus \langle v_i \rangle$.

$$A = \begin{pmatrix} 1 & 2 & \dots & n \\ n+1 & n+2 & & 2n \\ 2n+1 & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ n^2-n+1 & n^2-n+2 & & n^2 \end{pmatrix}$$

Since $A^{[2]} - A^{[1]} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1}$,

$kB + A^{[1]} = A^{[k+1]}$. And, $\{A^{[1]}, A^{[2]}\}$ linearly independent,
 then $\text{rank } A = 2$.

And, put $V_1 = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$, $V_2 = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ n \end{pmatrix}$. Then, for any $a_1, a_2 \in F$,

$$A(a_1 V_1 + a_2 V_2) = a_1 \begin{pmatrix} \sum_{k=1}^n k \\ n^2 + \sum_{k=1}^n k \\ 2n^2 + \sum_{k=1}^n k \\ \vdots \\ n^3 + \sum_{k=1}^n k \end{pmatrix} + a_2 \begin{pmatrix} \sum_{k=1}^n k^2 \\ n \cdot \left(\sum_{k=1}^n k\right) + \sum_{k=1}^n k^2 \\ 2n \cdot \left(\sum_{k=1}^n k\right) + \sum_{k=1}^n k^2 \\ \vdots \\ n^2 \cdot \left(\sum_{k=1}^n k\right) + \sum_{k=1}^n k^2 \end{pmatrix} = a_1 \begin{pmatrix} \frac{1}{2}n(n+1) \\ n^2 + \frac{1}{2}n(n+1) \\ 2n^2 + \frac{1}{2}n(n+1) \\ \vdots \\ n^3 + \frac{1}{2}n(n+1) \end{pmatrix} + a_2 \begin{pmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

$A[\langle V_1, V_2 \rangle]$ has rank 2.

$$AV_1 = n^2 V_2 + \sum_{k=1}^n k \cdot V_1$$

$$AV_2 = \left(\sum_{k=1}^n k^2\right) V_1 + \dots$$

$$A = \begin{pmatrix} 0 \cdot n \cdot V_1^t V_2^t \\ 1 \cdot n \cdot V_1^t + V_2^t \\ 2 \cdot n \cdot V_1^t + V_2^t \\ \vdots \\ (n-1) \cdot n \cdot V_1^t + V_2^t \end{pmatrix}$$

Inner product on Vector space V over F where $F = \mathbb{C}$ or $\mathbb{R}$ is a map:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F \quad \text{s.t.} \quad \begin{cases} \langle cx + y, z \rangle = c \cdot \langle x, z \rangle + \langle y, z \rangle \\ \overline{\langle x, y \rangle} = \langle y, x \rangle \\ x \neq 0 \Rightarrow \langle x, x \rangle > 0. \end{cases}$$

#6.1.9

If $x \neq 0$, then $\langle x, x \rangle = \sum_{i=1}^n \bar{a}_i \langle x, v_i \rangle = \sum 0 = 0$.

Contradiction.

$$\langle x, z \rangle = \langle y, z \rangle \Leftrightarrow \langle x - y, z \rangle = 0$$

$$\therefore x = y.$$

#6.1.10, 11, 12.

$$\begin{aligned} \|x + y\|^2 &= |\langle x + y, x + y \rangle| \\ &= |\langle x, x + y \rangle + \langle y, x + y \rangle| \\ &= |\langle x, x \rangle + \langle y, y \rangle| \\ &= \|x\|^2 + \|y\|^2. \end{aligned}$$

#6.1.15.

$$|\langle x, y \rangle| = \|x\| \cdot \|y\| \Leftrightarrow x = cy \text{ for some } c \in F.$$

$$\Leftrightarrow |\langle x, y \rangle| = |\langle x, x \rangle| \cdot |c| = |c| \cdot \|x\|^2$$

$$\|x\| \cdot \|y\| = |c| \cdot \|x\|^2.$$

$$\Rightarrow \text{Put } a = \frac{\langle x, y \rangle}{\|y\|^2}, \quad z = x - ay. \quad \text{Then,}$$

$$\langle z, y \rangle = \langle x - ay, y \rangle = \langle x, y \rangle - a \langle y, y \rangle = 0.$$

$$\text{Now, } \|x\|^2 = \|z + ay\|^2 = \|z\|^2 + a^2 \|y\|^2 = \|z\|^2 + \frac{\langle x, y \rangle^2}{\|y\|^4} \|y\|^2 = \|z\|^2 + \|x\|^2$$

$$\Rightarrow \|z\|^2 = 0$$

$$\text{Consequently, } x = ay.$$

#6.1.17.

$$\|x\|=0 \Leftrightarrow x=0$$

$$\text{Thus, } \ker T = \{0\} \quad | - |.$$

#6.1.18.

Let (V, F) , $(W, F, \langle \cdot, \cdot \rangle)$, $T: V \rightarrow W$ linear.

If $\langle T(x), T(y) \rangle$ is inner product of V , then

$$x \Rightarrow \text{if } \langle x, x \rangle = 0 \text{ if } \langle T(x), T(x) \rangle = 0 \text{ if } T(x) = 0$$

$$\text{Thus, } \ker T = \{0\}.$$

Conversely, if T is $| - |$, then

$$\begin{aligned} \langle x+y, z \rangle' &= \langle T(x+y), T(z) \rangle \\ &= \langle T(x) + T(y), T(z) \rangle \\ &= \langle T(x), T(z) \rangle + \langle T(y), T(z) \rangle \\ &= \langle x, z \rangle' + \langle y, z \rangle' \end{aligned}$$

$$\begin{aligned} \langle y, x \rangle' &= \langle T(y), T(x) \rangle \\ &= \overline{\langle T(x), T(y) \rangle} \\ &= \overline{\langle x, y \rangle} \end{aligned}$$

Positivity $\square$ (conv).

#6.1.19.

$$\begin{aligned}
 \|x+y\|^2 &= \langle x+y, x+y \rangle \\
 &= \langle x, x+y \rangle + \langle y, x+y \rangle \\
 &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &= \|x\|^2 + \|y\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} \\
 &= \|x\|^2 + \|y\|^2 + \operatorname{Re}(\langle x, y \rangle).
 \end{aligned}$$

$$\begin{cases} \|x\| = \|x-y+y\| \leq \|x-y\| + \|y\| \\ \|y\| \leq \|x-y\| + \|x\|. \end{cases} \Rightarrow \left| \|x\| - \|y\| \right| \leq \|x-y\|$$

#6.1.20.

$$\begin{aligned}
 i\|x+iy\|^2 &= i\|x\|^2 + i\|y\|^2 + \langle x, y \rangle - \overline{\langle x, y \rangle} \\
 -\|x-y\|^2 &= -\|x\|^2 - \|y\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle}. \\
 -i\|x-iy\|^2 &= -i\|x\|^2 - i\|y\|^2 + \langle x, y \rangle - \overline{\langle x, y \rangle} \\
 \|x+y\|^2 &= \|x\|^2 + \|y\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle}
 \end{aligned}$$

#6.1.21.

Let $A \in \mathcal{M}_n(\mathbb{C})$. Define

$$A_1 = \frac{1}{2}(A + A^*), \quad A_2 = \frac{1}{2i}(A - A^*).$$

Then, $A = A_1 + iA_2$ and $A_1^* = A_1, A_2^* = A_2$.

If $B_1, B_2 \in \mathcal{M}_n(\mathbb{C})$ satisfies $A = B_1 + iB_2, B_1^* = B_1, B_2^* = B_2$,

then $B_1 = A_1, B_2 = A_2$

proof.

$$B_1 = A - iB_2 = (A - iB_2)^* = A^* + iB_2,$$

$$B_2 = \frac{1}{2i}(A - A^*).$$

$$B_2 = \frac{1}{2i}(A - B_1) = -i(A - B_1) = (-i(A - B_1))^* = iA^* - iB_1,$$

$$\therefore 2iB_1 = iA + iA^*.$$

6, 1, 23.

$$\langle x, Ay \rangle = \left\langle x, \sum_{i=1}^n \left(\sum_{j=1}^n A_{i,j} \cdot y_j \right) \cdot e_i \right\rangle$$

$$= \sum_{i=1}^n \sum_{j=1}^n \overline{A_{i,j}} \cdot \overline{y_j} \cdot \langle x, e_i \rangle$$

$$= \sum_{i=1}^n \sum_{j=1}^n \overline{A_{i,j}} \cdot \overline{y_j} \cdot x_i$$

$$= \sum_{j=1}^n \sum_{i=1}^n \overline{A_{i,j}} \cdot x_i \cdot \overline{y_j}$$

$$= \sum_{j=1}^n \sum_{i=1}^n \overline{A_{i,j}} \cdot x_i \cdot \langle e_j, y \rangle$$

$$= \sum \sum \langle \overline{A_{i,j}} \cdot x_i \cdot e_j, y \rangle$$

$$= \langle \sum \sum \overline{A_{i,j}} \cdot x_i \cdot e_j, y \rangle = \langle A^* x, y \rangle$$

b). $\langle x, Ay \rangle = \langle A^* x, y \rangle = \langle \beta x, y \rangle \quad \forall x, y \in F^n$

$$\Rightarrow (A^* - \beta)x = 0 \quad \forall x \in F^n$$

$$\Rightarrow A^* - \beta = 0.$$

c). let $Q = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$. Then,

$$Q^* \cdot Q = \begin{pmatrix} \overline{v_1} \\ \vdots \\ \overline{v_n} \end{pmatrix} \cdot \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix} = I.$$

$$[T]_\beta = [I]_\alpha^\beta [T]_\alpha^\alpha [I]_\beta^\alpha$$

$$= Q^{-1} A Q$$

$$= Q^* A Q.$$

$$[U]_\beta = Q^{-1} A^* Q$$

$$= Q^* A^* Q.$$

6, 1, 29.
 $\bigvee$ over $\mathbb{R}$. with $\|\cdot\|$.

$$\langle x, y \rangle \stackrel{\text{def.}}{=} \frac{1}{4} \left(\|x+y\|^2 - \|x-y\|^2 \right) \quad \left(\begin{array}{l} \|x\|^2 + \|x-y\|^2 = 2\|x-y\|^2 + 2\|y\|^2. \\ \|x+y\|^2 + \|y\|^2 = 2\|x+y\|^2 + 2\|y\|^2. \end{array} \right)$$

$$\begin{aligned} \langle x, 2y \rangle &= \frac{1}{4} (\|x+2y\|^2 - \|x-2y\|^2) \\ &= \frac{1}{2} (\|x+y\|^2 - \|x-y\|^2) \\ &= 2\langle x, y \rangle. \end{aligned}$$

$$\|x+u+y\|^2 + \|x+u-y\|^2 = 2(\|x+u\|^2 + 2\|y\|^2)$$

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

$$\|u+y\|^2 + \|u-y\|^2 = 2(\|u\|^2 + \|y\|^2)$$

$$\begin{aligned} \langle x+u, 2y \rangle &= \frac{1}{4} (\|x+u+y\|^2 - \|x+u-y\|^2) \\ &= 2\langle x+u, y \rangle \stackrel{\text{WS}}{=} 2\langle x, y \rangle + 2\langle u, y \rangle \end{aligned}$$

$$\|x+u+y\|^2 + \|x+u\|^2 =$$

Lemma. Let $S = \{v_1, \dots, v_k\}$ be a ^{non-zero} orthogonal vectors set.

Then, for any $y \in \langle S \rangle$,

$$y = \sum_{i=1}^k \frac{\langle y, v_i \rangle}{\|v_i\|^2} v_i.$$

Proof. Denote $y = \sum_{i=1}^k \alpha_i v_i$ ($\alpha_i \in \mathbb{F}$). Then, for each $1 \leq j \leq k$,

$$\langle y, v_j \rangle = \left\langle \sum_{i=1}^k \alpha_i v_i, v_j \right\rangle = \sum_{i=1}^k \alpha_i \langle v_i, v_j \rangle = \alpha_j \langle v_j, v_j \rangle = \alpha_j \|v_j\|^2.$$

$$\therefore \alpha_j = \frac{\langle y, v_j \rangle}{\|v_j\|^2}.$$

Corollary,

Non-zero orthogonal vectors set is linearly indep.

Gram-Schmidt process.

Let $S = \{w_1, \dots, w_n\}$ be a linearly indep. set.

Then, $S' = \{v_1, \dots, v_n\}$ is an orthogonal set s.t. $\langle S \rangle = \langle S' \rangle$,

where

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j.$$

Thm. Finite vector space has a orthonormal basis $\beta = \{v_1, \dots, v_n\}$.

In particular, for any $x \in V$,

$$x = \sum_{i=1}^n \langle x, v_i \rangle \cdot v_i.$$

Corollary, $A = [T]_{\beta}$ then $A_{ij} = \langle T(v_j), v_i \rangle$.

Def. $S \subseteq V$.

$$S^\perp = \{x \in V \mid \forall y \in S, \langle x, y \rangle = 0\}.$$

Thm. Let $W \subseteq V$. Then, $V = W \oplus W^\perp$.

$$y = u + z$$

$$= \sum_{i=1}^k \langle y, u_i \rangle u_i$$

($\{u_1, \dots, u_k\}$ is orthonormal basis for W)

6, 2, 6.

Let $x \in W$ be given. Then, $x = z + y$, $\exists z \in W, y \in W^\perp$.

$$\langle x, y \rangle = \langle z + y, y \rangle = \|y\|^2.$$

If $\|y\|^2 = 0$, then $y = 0$. But, this implies $x = z \in W$.

6, 2, 10.

T: $V \rightarrow V$; $x_1 + x_2 \mapsto x_1$

$$\|x_1 + x_2\|^2 = \|x_1\|^2 + \|x_2\|^2 \geq \|x_1\|^2$$

$$\Rightarrow \|x\| \geq \|Tx\|.$$

6.2.12. $A \in M_{m \times n}(F)$ $L_A: F^m \rightarrow F^n$.

Let $x \in N(L_A)$ be given. Then is, $Ax = 0$.

$$\begin{aligned} \forall y \in R(L_A), \langle x, y \rangle &= \langle x, A^T z \rangle \text{ for some } z \in F^n \\ &= \langle A x, z \rangle \\ &= 0. \end{aligned}$$

$$\therefore x \in (R(L_A))^{\perp}.$$

Conversely, let $x \in (R(L_A))^{\perp}$.

Then is, $\langle x, A^T z \rangle = 0$ for all $z \in F^n$.

$$\therefore Ax = 0.$$

13. $V, W \subseteq V, \dim W = k$.

$$1) S_0 \subseteq S \Rightarrow S^{\perp} \subseteq S_0^{\perp}$$

$$\begin{aligned} \text{Let } x \in S^{\perp}. \forall y \in S, \langle x, y \rangle &= 0. \\ &\Rightarrow \forall y \in S_0, \langle x, y \rangle = 0 \\ &\Rightarrow x \in S_0^{\perp}. \end{aligned}$$

$$\begin{aligned} 2) \text{ Let } x \in S. \text{ Then, for any } y \in S^{\perp}, \\ \langle x, y \rangle &= 0. \\ &\Rightarrow x \in (S^{\perp})^{\perp}. \end{aligned}$$

$$\begin{aligned} 3) \text{ If } x \notin W, \text{ then } \exists y \in W^{\perp} \text{ s.t. } \langle x, y \rangle &\neq 0. \\ \text{Then is, } x &\notin (W^{\perp})^{\perp}. \end{aligned}$$

d

Lemma. Let V be an inner product space.

Suppose $S \stackrel{\text{def}}{=} \{v_1, v_2, \dots, v_k\}$ is a subset of V which is consisted non-zero orthogonal,

Then, for any $y \in \langle S \rangle$,

$$y = \sum_{i=1}^k \frac{\langle y, v_i \rangle}{\|v_i\|^2} \cdot v_i$$

Proof. Denote $y = \sum_{i=1}^k \alpha_i v_i$. Then, for each $1 \leq j \leq k$,

$$\langle y, v_j \rangle = \left\langle \sum_{i=1}^k \alpha_i v_i, v_j \right\rangle = \sum_{i=1}^k \alpha_i \langle v_i, v_j \rangle = \alpha_j \langle v_j, v_j \rangle = \alpha_j \|v_j\|^2.$$

$$\text{Now, } \alpha_j = \frac{\langle y, v_j \rangle}{\|v_j\|^2}$$

Corollary, orthogonal subset is linearly independent.

Gram-Schmidt process.

Let V be an inner product space, and

$S = \{w_1, \dots, w_n\} \subseteq V$ be a linearly independent subset.

Define a set $S' = \{v_1, \dots, v_n\}$ as, for each $1 \leq k \leq n$,

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} \cdot v_j$$

Then, S' is orthogonal set satisfying $\langle S \rangle = \langle S' \rangle$.

Corollary, Every finite vector space has an orthonormal basis β .

Especially, if $\beta = \{v_1, \dots, v_n\}$, then for any $x \in V$,

$$x = \sum_{i=1}^n \langle x, v_i \rangle \cdot v_i.$$

And for any linear operator $T: V \rightarrow V$,

$$([T]_{\beta})_{ij} = \langle T(v_j), v_i \rangle.$$



Def. Let $V = F[x]$ be a polynomial space with a field $F = \mathbb{R}$ or $F = \mathbb{C}$.
 Define a map:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F : \left(\sum_{i=0}^n a_i x^i, \sum_{i=0}^m b_i x^i \right) \mapsto \sum_{i=0}^{\infty} a_i \overline{b_i}.$$

well-definedness is given by coefficient of $p(x) \in V$ is all zero but finite.
 And, it is satisfied Inner product Axiom: Conjugate and positiveness are clear.
 Sesquilinearity too.

Moreover, $\beta = \{x^n \mid n=0,1,\dots\}$ is an orthonormal basis for V .

Consider a subspace $W = \langle \{1+x^n \mid n \in \mathbb{N}\} \rangle$.

Then, $1 \notin W$ clearly. That is, $W \neq V$.

Furthermore, $W^\perp = \{0\}$ because: let $p(x) \in W^\perp$.

Then, for any $x^n \in \beta$, $\langle p(x), x^n \rangle = 0$. This implies $|p(x)| = 0$.

Consequently, $(W^\perp)^\perp = (\{0\})^\perp = V \neq W$.

Lemma. Let β be an orthonormal basis, α be a standard basis for F^n . ($F = \mathbb{R}$ or $\mathbb{C}$)
 Then, $([I]_{\alpha}^{\beta})^* = ([I]_{\beta}^{\alpha})^{-1} = [I]_{\beta}^{\alpha}$

Def. Let V be a fin. dim Inner Product Space, $T: V \rightarrow V$ be a linear operation.
 A linear operator T^* is called adjoint of T if:
 For any orthonormal basis β , $[T^*]_{\beta} = [T]_{\beta}^*$.

Lemma. Let V be a fin. dim. Inner product space.

If $g: V \rightarrow F$ is a linear map, then there exists uniquely $y \in V$ s.t. $g(x) = \langle x, y \rangle$.
 Proof. Let $\beta = \{v_1, \dots, v_n\}$ be an orthonormal basis for V . Put

$$y = \sum_{i=1}^n \overline{g(v_i)} \cdot v_i.$$

Clearly, a map $V \rightarrow F: x \mapsto \langle x, y \rangle$ is linear. Moreover, for all $1 \leq i \leq n$,

$$\langle v_j, y \rangle = \sum_{i=1}^n g(v_i) \cdot \langle v_j, v_i \rangle = \sum_{i=1}^n g(v_i) \cdot \delta_{ji} = g(v_j)$$

Uniqueness is given by:

$$\forall x \in V, \langle x, y \rangle = \langle x, y' \rangle \Rightarrow y = y'.$$

Thm. Let V be a f.i. p.v.s. $T: V \rightarrow V$ be a linear map. Then, there exists one and only one map $T^*: V \rightarrow V$ s.t.

$$\text{For any } x, y \in V, \langle T(x), y \rangle = \langle x, T^*(y) \rangle.$$

Specifically, T^* is linear.

proof. Define a map

$$g: V \rightarrow F: x \mapsto \langle T(x), y \rangle.$$

Clearly, g is linear.

Since above lemma, there exists a unique $y^* \in V$ s.t. for any $x \in V$,

$$g(x) = \langle T(x), y \rangle = \langle x, y^* \rangle$$

Now, define $T^*: V \rightarrow V: y \mapsto y^*$.

To show that T^* is linear, let $x, y_1, y_2 \in V$, $\alpha \in F$ be given.

$$\begin{aligned} \langle x, T^*(\alpha y_1 + y_2) \rangle &= \langle T(x), \alpha y_1 + y_2 \rangle \\ &= \alpha \langle T(x), y_1 \rangle + \langle T(x), y_2 \rangle \\ &= \alpha \langle x, T^*(y_1) \rangle + \langle x, T^*(y_2) \rangle \\ &= \langle x, \alpha T^*(y_1) + T^*(y_2) \rangle \end{aligned}$$

Since x was given arbitrarily,

$$T^*(\alpha y_1 + y_2) = \alpha T^*(y_1) + T^*(y_2).$$

Similarly, the uniqueness of T^* is shown.

Def. Let V, W be Inner product spaces (not necessary finite).

For a linear map $T: V \rightarrow W$ if exists, $T^*: V \rightarrow V$ is called adjoint of T if:

$$\forall x \in V, y \in W, \langle T(x), y \rangle_W = \langle x, T^*(y) \rangle_V$$

Thm. Let V be a f.i. inn. pro. space and β be an orthonormal basis,

For a linear operator $T: V \rightarrow V$,

$$[T^*]_{\beta} = [T]_{\beta}^{\#}.$$

Least squares approximation

Lemma. Let $A \in M_{mn}(F)$ be a Matrix.

1) For any $x \in F^n, y \in F^m$; $\langle Ax, y \rangle_m = \langle x, A^* y \rangle_n$.

2) $\text{rank}(A^* A) = \text{rank}(A)$.

3) $\text{rank } A = n \iff A^* A$ is invertible.

proof. Show there 1),

$$\langle Ax, y \rangle_m = y^* \cdot (Ax) = (A^* y)^* x = \langle x, A^* y \rangle_n$$

Show 2). $Ax=0 \Rightarrow A^* Ax=0$ is clear.

If $A^* Ax=0$, then

$$0 = \langle A^* Ax, x \rangle_n = \langle Ax, Ax \rangle_n = \|Ax\|^2$$

Thus, $Ax=0$. In sum, $\ker A = \ker A^* A$. This implies

$$\text{rank } A^* A = n - \dim \ker A = n - \dim \ker A^* A = \text{rank } A.$$

Thm. Let $A \in M_{mn}(F), y \in F^m$ be given. Then, there exists $x_0 \in F^n$ s.t.

$$\text{For all } x \in F^n, \|Ax_0 - y\| \leq \|Ax - y\|.$$

In particular, if $\text{rank } A = n$, then x_0 is uniquely determined as:

$$x_0 = (A^* A)^{-1} A^* y.$$

6.3.7.

Let $T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, then $T^* = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

$$\ker T = \left\{ a \begin{pmatrix} 1 \\ -1 \end{pmatrix} \mid a \in F \right\}$$

$$\ker T^* = \left\{ a \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mid a \in F \right\}.$$

Thm. Let V be a fin. dim. Inn. space, and $T: V \rightarrow V$ be a linear operator. Then
 T is invertible iff T^* is invertible

In particular, $(T^*)^{-1} = (T^{-1})^*$.

Proof. Let β be an orthonormal basis for V .

If T is invertible, then there exists inverse of a matrix representation s.t.

$$[T]_{\beta}^{-1} \cdot [T]_{\beta} = I.$$

Since $I^* = I$,

$$I = ([T]_{\beta})^* \cdot ([T]_{\beta}^{-1})^* = [T^*]_{\beta} \cdot ([T]_{\beta}^{-1})^*.$$

That is,

$$[T^*]_{\beta}^{-1} = ([T]_{\beta}^{-1})^* = ([T^{-1}]_{\beta})^* = [T^{-1}]_{\beta}^*.$$

Thm. If a Inn. prod. space V can be represented as $V = W \oplus W^{\perp}$, then

for a projection $T: V \rightarrow V; x_1 + x_2 \mapsto x_1$, $T = T^*$.

Proof. Let $x_1 + x_2, y_1 + y_2 \in V$ ($x_1, y_1 \in W$, $x_2, y_2 \in W^{\perp}$) be given. Then,

$$\langle T(x_1 + x_2), y_1 + y_2 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle = \langle x_1, T(y_1 + y_2) \rangle = \langle x_1 + x_2, T(y_1 + y_2) \rangle$$

Thm. Let V be an Inn. prod. space, $T: V \rightarrow V$ be a linear operator. Then,

$$\forall x \in V, \|T(x)\| = \|x\| \text{ iff } \forall x, y \in V, \langle T(x), T(y) \rangle = \langle x, y \rangle$$

Proof. Converse is clear. Let $x, y \in V$ be given. Then,

$$\begin{aligned} \langle T(x), T(y) \rangle &= \frac{1}{4} \cdot (\|iT(x) + T(y)\|^2 - \|T(x) - T(y)\|^2 - i\|T(x) - iT(y)\|^2 + \|T(x) + T(y)\|^2) \\ &= \frac{1}{4} \cdot (i\|x + iy\|^2 - \|x - y\|^2 - i\|x - iy\|^2 + \|x + y\|^2) \\ &= \langle x, y \rangle. \end{aligned}$$

6.3.11. If $T^*T = T_0$, then $T = T_0$.

$\forall x \in V$, $T^*T(x) = 0$. That is, for any $x \in V$,

$$0 = \langle T^*T(x), x \rangle = \langle T(x), T(x) \rangle = \|T(x)\|^2$$

This implies $T(x) = 0$.

6.3.12. Let V be a Inn. pro. space, $T: V \rightarrow V$ be a linear.

If exists, $R(T^*)^\perp = N(T)$

Proof. Let $x \in R(T^*)^\perp$ be given. Then, for any $y \in V$,

$$0 = \langle x, T^*(y) \rangle = \langle T(x), y \rangle$$

Since y was chosen arbitrarily, $T(x) = 0$. $\therefore x \in N(T)$.

Conversely, let $x \in N(T)$. Then, for any $y \in V$,

$$0 = \langle T(x), y \rangle = \langle x, T^*(y) \rangle.$$

$\therefore x \in R(T^*)^\perp$.

6.3.13. Let V be a fin. dim. Inn. space and $T: V \rightarrow V$ be a linear operation.

$$1). \ker T^*T = \ker T$$

$$2). \text{rank } T = \text{rank } T^* = \text{rank } T^*T = \text{rank } T^*T.$$

Proof. Trivially, $\ker T \leq \ker T^*T$.

Let $x \in \ker T^*T$. Then,

$$0 = \langle T^*T(x), x \rangle = \langle T(x), T(x) \rangle = \|T(x)\|^2$$

Thus, $T(x) = 0$. Moreover,

$$\text{Im } T \cong V / \ker T = V / \ker T^*T \cong \text{Im } T^*T.$$

Since $R(T^*)^\perp = N(T)$, $\text{rank } T^* = n - \text{nullity}(T) = \text{rank } T$.

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Not necessary finite

Lemma. Let V be a T_m space, and $Y, Z \in V$ be vectors. Define

$$T: V \rightarrow V: \langle x, Y \rangle \cdot Z$$

Then, T is linear and an Adjoint of T exists. Especially, $T^*: V \rightarrow V: x \mapsto \overline{\langle Z, x \rangle} \cdot Y$

Def. Let V, W be fin. dim. prod. spaces β, γ be orthonormal basis for V, W respectively.

For a linear map $T: V \rightarrow W$,

1) Adjoint T^* of T exists uniquely, and T^* is linear.

$$2) [T^*]_{\beta}^{\gamma} = ([T]_{\beta}^{\gamma})^*$$

$$3) \text{rank } T = \text{rank } T^*$$

$$4) \text{ For all } x, y \in V, \langle T^*x, y \rangle_V = \langle x, T(y) \rangle_W.$$

$$5) T^*T(x) = 0 \quad \text{iff} \quad T(x) = 0.$$

Proof. Let $y \in W$ be fixed. Define

$$\varphi: V \rightarrow F; x \mapsto \langle T(x), y \rangle_W$$

Then, clearly, φ is a linear functional. By Lemma, there uniquely exists $u \in V$ s.t.

$$\forall x \in V, \langle T(x), y \rangle_W = \langle x, u \rangle_V$$

Now, define adjoint of T as:

$$T^*: W \rightarrow V; y \mapsto u.$$

Let $V = F[x]$, $\beta = \{x^i \mid i=0,1,\dots\}$, Define a map:

$$T: V \rightarrow V: \sum_{i=0}^{\infty} a_i x^i \mapsto \sum_{k=0}^{\infty} \left(\sum_{i=k}^{\infty} a_i \right) x^k.$$

Linearity is clear. For any $n=0,1,2,\dots$,

$$T(x^n) = 1 + x + x^2 + \dots + x^n = \sum_{i=0}^n x^i.$$

Now, suppose that there exists an adjoint $T^\sharp$.

That is, $T^\sharp: V \rightarrow V$ is a linear map s.t

$$\text{For any } \sum a_i x^i, \sum b_i x^i \in V, \langle T(\sum a_i x^i), \sum b_i x^i \rangle = \langle \sum a_i x^i, T^\sharp(\sum b_i x^i) \rangle$$

Now, for any $n=0,1,2,\dots$, and $p(x) \in V$,

$$\begin{aligned} \langle T(x^n), p(x) \rangle &= \langle x^n, T^\sharp(p(x)) \rangle \\ &= \left\langle \sum_{i=0}^n x^i, p(x) \right\rangle \\ &= \sum_{i=0}^n a_i \quad \left(\text{where } p(x) = \sum_{i=0}^{\infty} a_i x^i \right) \end{aligned}$$

Let $p(x) = 1$. Then, for any $n \geq 1$,

$$\begin{aligned} \langle x^n, T^\sharp(1) \rangle &= b_n \quad \left(\text{where } T^\sharp(1) = \sum_{i=0}^{\infty} b_i x^i \right) \\ &= 1. \end{aligned}$$

This is contradiction because $T^\sharp(1) \notin V$.

$$\begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}}_a + \underbrace{\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}}_b$$

$$b^2 = 0.$$

$$(a+b)^{2025} - b = a.$$

$$ab = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(a+b)^{2025}$$

$$= \sum_{i=0}^{2025} \binom{2025}{i} \cdot a^i \cdot b^{2025-i}$$

$$= a^{2025} + \cancel{2025 \cdot a^{2024} \cdot b} + 2025 \cdot a^{2024} \cdot b.$$

$$= a^{2025} + 2025 \cdot a^{2024} \cdot b$$

$$a = \begin{bmatrix} 2^{\frac{1}{2025}} & 0 \\ 0 & 5^{\frac{1}{2025}} \end{bmatrix}$$

$$2025 \cdot \begin{bmatrix} 2^{\frac{4}{2025}} & 0 \\ 0 & 5^{\frac{4}{2025}} \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ r & 0 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 & 0 \\ r & 0 \end{bmatrix} = 2025 \cdot \begin{bmatrix} 0 & 0 \\ r \cdot 5^{\frac{4}{2025}} & 0 \end{bmatrix} = z$$

$$= \begin{bmatrix} 0 & 0 \\ \frac{2}{2025} \cdot \frac{1}{5^{\frac{4}{2025}}} & 0 \end{bmatrix}$$

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^4 : (x_1, x_2, x_3, x_4) \mapsto x_1 + x_2 - x_3 - x_4$$

$$\ker T = W.$$

$$T = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

$$(\ker T)^\perp = \text{Im}(T^*) = \langle \{1, 1, -1, -1\} \rangle$$

$$(1 \ 1 \ -1 \ -1) \cdot x$$

$$= x \quad x \quad -x \quad -x$$

$$\begin{pmatrix} \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4} \end{pmatrix} + \begin{pmatrix} \frac{3}{4}, \frac{3}{4}, \frac{1}{4}, \frac{5}{4} \end{pmatrix}$$

$$(1, 1, -1, -1) + (0, 0, 1, 2) \in W^\perp$$

$$(a, a, -a, -a) \neq (x_1, x_2, x_3, x_4) = a$$

$$x_1 + x_2 = x_3 + x_4$$

$$(a, a, -a, -a) + (x_1, x_2, x_3, x_4 + x_2 - x_3)$$

$$(x_1, x_1, x_3, 2x_1 - x_3)$$

$$\begin{cases} a + x_1 = 1 \\ -a + x_3 = 0 \\ 2x_1 - x_3 - a = 1 \end{cases}$$

$$2x_1 - 2a = 1$$

$$x_1 - a = \frac{1}{2}$$

$$x_1 + a = 1$$

$$2x_1 = \frac{3}{2}$$

$$x_1 = \frac{3}{4}$$

$$a = \frac{1}{4}$$

0

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